

No. of Functional Features Included in the Solution

Date	06 July 2026
Project Name	Credit Card Approval Prediction System
Prepared By	Joyal Kumar, Keerthi, Yasawini, Nasir, Harika

Lists the functional features actually present in the Credit Card Approval Prediction repository, based on a direct read-through of main.py and app.py, with implementation status and relative contribution to the overall solution.

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Data Cleaning & Preprocessing	Removes duplicate applicants and fills missing OCCUPATION_TYPE values before modeling	main.py	Done	High
2	Target Label Derivation	Converts monthly credit STATUS history into a single Good/Bad target per customer via grouping	main.py	Done	High
3	Categorical Feature Encoding	Label-encodes 8 categorical columns (gender, education, income type, etc.) for model input	main.py	Done	High
4	Multi-Model Training & Comparison	Trains and evaluates Logistic Regression, Decision Tree, and Random Forest, reporting accuracy, F1 score, and confusion matrix for each	main.py	Done	High
5	Model Persistence	Saves the selected Random Forest model to model.pkl via pickle for reuse without retraining	main.py	Done	Medium
6	Applicant Input Form	Web form collecting 17 applicant fields (income, education, family status, employment, etc.)	templates/index.html	Done	High
7	Prediction Endpoint	Flask POST /predict route that builds a feature vector from form data and returns a Good/Bad Customer prediction	app.py	Done	High
8	Result Display Page	Shows the applicant's name and prediction outcome on a dedicated results page	templates/result.html	Done	Medium
9	Basic Error Handling	Wraps the /predict route in a try/except, though it currently surfaces raw exception text to the user rather than a friendly message	app.py	In Progress	Low
10	Consistent Encoder Reuse at Inference	Persisting and reusing the same LabelEncoder from training inside app.py, so categorical codes can't drift between the two	main.py / app.py	Pending	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	8 Done, 1 In Progress, 1 Pending
Core / Must-Have Features	7 (Data cleaning, target derivation, encoding, model training/comparison, input form, prediction endpoint, result display)
Additional / Nice-to-Have Features	3 (Model persistence, error handling, encoder consistency at inference)
Features Tested & Verified	7 of 10 - verified by cloning the repo and running the app end-to-end (see Coding & Solution and Performance Testing documents); encoder consistency was not tested since it is not yet implemented

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Input Form, Result Display Page
2	Backend / Logic	5	Data Cleaning, Target Label Derivation, Categorical Encoding, Multi-Model Training & Comparison, Prediction Endpoint
3	Database / Storage	1	Model Persistence (model.pkl) - no live database is used
4	API / Integration	0	None - no external/third-party APIs are called
5	Security / Authentication	0	None currently implemented - no login or access control exists